SECTION 1

## Cloning the Repository

- **GitHub source:** Samples live in the AI-Agent-Workflows repository organized by chapter folder under a single project root.
- **Clone workflow:** A standard `git clone` followed by `cd` into the repo places all listings where subsequent setup commands expect them.
- **First-time order:** Cloning is the opening step; later sections assume commands run from that repository root.

SECTION 2

## Creating a Python Environment

- **Version requirement:** The book targets Python 3.11 or newer, with a dedicated virtual environment recommended over the system interpreter.
- **Platform activation:** Windows uses `python -m venv venv` and `venv\Scripts\activate`; macOS and Linux use `python3 -m venv venv` and `source venv/bin/activate`.
- **Interpreter selection:** VS Code users can pick a project interpreter via the command palette so the editor matches the environment used in the terminal.

SECTION 3

## Installing Dependencies and Configuring the API Key

- **Dependency install paths:** Pressing F5 in VS Code can auto-install requirements, or `pip install -r requirements.txt` installs pinned packages manually inside the active environment.
- **Secrets file:** Copy `.env.example` to `.env` and set `OPENAI_API_KEY` with a key from the OpenAI platform; never commit the `.env` file because it holds billing credentials.
- **Pinned requirements:** The root `requirements.txt` locks versions the chapter scripts were tested against, reducing “works on my machine” drift.

SECTION 4

## Running Samples and Troubleshooting

- **Running a sample:** Execute a chapter script from the repo root, e.g. `python chapter_02/01_first_agent.py`, mirroring the path pattern for other chapters.
- **Authentication failures:** Usually mean a missing `.env`, a wrong working directory, or a rotated key that was not updated locally.
- **Import failures:** Typically indicate the virtual environment is inactive or packages were installed into a different Python than the one running the script.
- **Healthy habits:** Keep interpreter, venv, and `.env` aligned, rely on `.gitignore` for secrets, and refresh the key in `.env` after any credential rotation.